Group C: Blockchain Technology

Laboratory Practice III – Oral / Viva Questions with Short Answers

SPPU – Fourth Year Computer Engineering (2019 Course)

Prepared from the Group C assignment list and the Blockchain Technology topics in the syllabus you shared. The syllabus specifies any five Group C assignments and one mini-project as mandatory.

1. Blockchain Fundamentals

Q1. What is Blockchain?

Answer: A distributed digital ledger that stores records in blocks linked together using cryptographic hashes.

Q2. Why is it called a blockchain?

Answer: Because blocks of records are connected in sequence using cryptographic links.

Q3. What is a block?

Answer: A data structure that contains a set of transactions or records, along with metadata such as a timestamp and the previous block's hash.

Q4. What is a distributed ledger?

Answer: A record of transactions that is shared and maintained across multiple participating computers.

Q5. What is decentralization?

Answer: Control and record keeping are distributed among multiple participants instead of being handled by one central authority.

Q6. What is the difference between centralized and decentralized systems?

Answer: A centralized system is controlled by a central authority; a decentralized system distributes control among participants.

Q7. What is a hash function?

Answer: A function that converts input data into a fixed-length output called a hash.

Q8. What is SHA-256?

Answer: A cryptographic hash function that produces a 256-bit hash value.

Q9. Can a hash be used to directly recover the original data?

Answer: No. A cryptographic hash function is designed to be one-way; it is not encryption.

Q10. What is immutability in Blockchain?

Answer: It means recorded data is difficult to alter without detection, especially after it has been accepted by the network.

Q11. What is a node?

Answer: A computer participating in a blockchain network by maintaining data, validating information, or communicating with other nodes.

Q12. What is a transaction?

Answer: A recorded operation or transfer submitted to a blockchain network.

2. Cryptography, Digital Signatures and Merkle Trees

Q1. What is cryptography?

Answer: The use of mathematical techniques to protect information and support confidentiality, integrity and authentication.

Q2. What is symmetric-key cryptography?

Answer: A method in which the same secret key is used for encryption and decryption.

Q3. What is asymmetric-key cryptography?

Answer: A method that uses a related public key and private key pair.

Q4. What is the main difference between symmetric and asymmetric cryptography?

Answer: Symmetric cryptography uses one shared secret key; asymmetric cryptography uses a public/private key pair.

Q5. What is a public key?

Answer: A key that can be shared publicly and is used in operations such as verifying digital signatures or encrypting data in suitable schemes.

Q6. What is a private key?

Answer: A secret key controlled by its owner and used for operations such as creating digital signatures.

Q7. What is a digital signature?

Answer: A cryptographic mechanism that helps verify the authenticity and integrity of a digitally signed message or transaction.

Q8. Does a digital signature provide confidentiality by itself?

Answer: No. It provides authenticity and integrity properties, not confidentiality by itself.

Q9. What is a Merkle Tree?

Answer: A tree structure in which leaf nodes contain hashes of data and parent nodes contain hashes derived from their child hashes.

Q10. What is a Merkle Root?

Answer: The top hash of a Merkle Tree, representing the data included in that tree.

Q11. Why are Merkle Trees useful in Blockchain?

Answer: They allow efficient verification that a transaction belongs to a set of transactions without needing to inspect all the data.

3. Blockchain Platforms and Consensus Algorithms

Q1. Name blockchain platforms listed in the syllabus.

Answer: Bitcoin, Ethereum, Hyperledger, IOTA, Corda and R3.

Q2. What are the three blockchain platform types listed in the syllabus?

Answer: Public, private and consortium blockchains.

Q3. What is a public blockchain?

Answer: A blockchain that is generally open for anyone to participate in according to its network rules.

Q4. What is a private blockchain?

Answer: A blockchain in which participation and access are restricted by an organisation or administrator.

Q5. What is a consortium blockchain?

Answer: A blockchain governed by a group of selected organisations rather than one organisation alone or the public at large.

Q6. What is consensus in Blockchain?

Answer: A process by which participating nodes agree on the accepted state or order of records in the network.

Q7. Why is a consensus algorithm needed?

Answer: It helps distributed participants agree on valid records without relying on one central authority.

Q8. What is Proof of Work (PoW)?

Answer: A consensus mechanism in which participants perform computational work to propose or validate blocks under the protocol rules.

Q9. What is Proof of Stake (PoS)?

Answer: A consensus mechanism in which validators are selected or weighted based on stake committed to the network.

Q10. What is the Byzantine Generals Problem?

Answer: A problem that describes how distributed participants can reach agreement even when some participants may be faulty or malicious.

Q11. What is Proof of Elapsed Time?

Answer: A consensus approach that uses assigned wait times and trusted execution mechanisms to select a participant to create a block.

Q12. What is Proof of Burn?

Answer: A consensus approach in which participants irreversibly destroy or spend tokens to demonstrate commitment.

Q13. What is Proof of Activity?

Answer: A hybrid consensus approach combining elements of Proof of Work and Proof of Stake.

4. Cryptocurrency, Bitcoin and Wallets

Q1. What is cryptocurrency?

Answer: A digital asset that uses cryptographic techniques and a blockchain or other distributed ledger for its operation.

Q2. What is Bitcoin?

Answer: A cryptocurrency that operates on a peer-to-peer blockchain network.

Q3. What is a cryptocurrency wallet?

Answer: A tool that manages the keys used to access and authorise transactions involving blockchain assets.

Q4. Does a wallet usually store coins physically?

Answer: No. Assets are recorded on the blockchain; the wallet manages the keys used to control them.

Q5. What is MetaMask?

Answer: A cryptocurrency wallet commonly used to interact with Ethereum-compatible blockchain networks and decentralised applications.

Q6. What is a wallet address?

Answer: A public identifier used to receive digital assets or interact with a blockchain account.

Q7. What is a seed phrase?

Answer: A sequence of words used to recover a wallet's keys; it must be kept secret.

Q8. Why should a private key or seed phrase never be shared?

Answer: Anyone who obtains it may be able to control the associated wallet and assets.

Q9. What is Ether (ETH)?

Answer: The native cryptocurrency of the Ethereum network.

Q10. What is a transaction fee?

Answer: A fee paid to have a transaction processed by the blockchain network.

Q11. What is a test network?

Answer: A blockchain network used for development and testing, typically with test assets that have no intended real-world monetary value.

5. Ethereum, Smart Contracts and Solidity

Q1. What is Ethereum?

Answer: A blockchain platform that supports programmable smart contracts and decentralised applications.

Q2. What is a smart contract?

Answer: A program deployed on a blockchain that executes according to its coded rules when invoked.

Q3. What is Solidity?

Answer: A programming language commonly used to write smart contracts for the Ethereum Virtual Machine.

Q4. What is the Ethereum Virtual Machine (EVM)?

Answer: The execution environment that runs compatible smart contract bytecode on Ethereum and EVM-compatible networks.

Q5. What is a decentralised application (dApp)?

Answer: An application that uses smart contracts or other decentralised infrastructure for some of its functionality.

Q6. What does deploying a smart contract mean?

Answer: Publishing the compiled contract to a blockchain so it receives an address and can be called.

Q7. What is a constructor in Solidity?

Answer: A special function that runs once when a contract is created and is commonly used to initialise state.

Q8. What is a struct in Solidity?

Answer: A user-defined data type that groups related variables under one name.

Q9. What is an array in Solidity?

Answer: A collection of elements of the same declared type, accessed using indexes.

Q10. What is a fallback function?

Answer: A special function that can be triggered when a call does not match another function signature, or under specified receive/fallback conditions.

Q11. What is gas in Ethereum?

Answer: A unit used to measure the computational work required to execute operations on the EVM.

Q12. Why does a smart contract transaction consume gas?

Answer: Because executing contract operations uses network computational resources.

Q13. What is the difference between gas and Ether?

Answer: Gas measures computational work; Ether is the asset used to pay transaction execution fees on Ethereum.

Q14. What are Swarm and Whisper as mentioned in the syllabus?

Answer: Swarm is described as a decentralised storage platform, while Whisper is described as a decentralised messaging platform.

6. Blockchain Applications and Use Cases

Q1. Name application areas listed in the syllabus.

Answer: Retail, banking and financial services, government, healthcare, IoT, energy and utilities.

Q2. How can Blockchain be used in banking?

Answer: It can support shared transaction records and audit trails among participating institutions.

Q3. How can Blockchain be used in healthcare?

Answer: It can support controlled sharing and auditability of records, subject to privacy and access-control design.

Q4. How can Blockchain be used in supply chains or retail?

Answer: It can help participants record and trace product movements and related events.

Q5. What is a blockchain-based e-voting system?

Answer: A voting application that records voting events or results using blockchain mechanisms to support auditability and tamper resistance.

Q6. What is a blockchain-based charity application?

Answer: An application that records donations and fund movements to make transactions more transparent to permitted participants.

Q7. What is a key limitation to consider before using Blockchain?

Answer: Blockchain may add complexity, cost and performance overhead; it is not necessary for every application.

Q8. Does Blockchain automatically make all stored data private?

Answer: No. Privacy depends on the network design, permissions, encryption and what data is stored on-chain.

7. Practical and Viva Questions Based on Group C Assignments

Q1. What is the purpose of installing MetaMask in the practical?

Answer: To create or access a wallet and interact with a supported blockchain network.

Q2. What should you check when studying Ether spent per transaction?

Answer: The transaction details, gas used, gas price or fee information, and the network on which it occurred.

Q3. What are the basic operations in the bank-account smart contract practical?

Answer: Deposit money, withdraw money and show the account balance.

Q4. Why should a bank smart contract check the withdrawal amount?

Answer: To prevent invalid withdrawals, such as withdrawing more than the available balance.

Q5. What data might be stored in the Student smart contract practical?

Answer: Student-related fields defined by the program, organised using Solidity structures and arrays.

Q6. Why observe transaction fee and gas values after deployment or execution?

Answer: To understand the computational cost of contract operations on the selected network.

Q7. What is Hyperledger?

Answer: An open-source set of enterprise blockchain technologies and frameworks.

Q8. What is a business network in the Hyperledger practical?

Answer: A network of participants, assets and transactions modelled for a business use case.

Q9. What is the difference between a blockchain survey report and a smart contract practical?

Answer: A survey report studies blockchain types and real-world uses; a smart contract practical implements executable blockchain logic.

Q10. Name the mini-project topics listed in Group C.

Answer: Blockchain-based e-voting, transparent and genuine charity, health-related medical records, and mental health applications.

Quick Revision

SHA-256: Cryptographic hash function producing a 256-bit hash.
Symmetric cryptography: Same secret key for encryption and decryption.
Asymmetric cryptography: Public/private key pair.
Consensus: Agreement on the accepted ledger state.
PoW: Consensus based on computational work.
PoS: Consensus based on stake.
Smart contract: Program deployed on a blockchain.
Solidity: Language for writing EVM smart contracts.
Gas: Measure of computational work on Ethereum.

Practical exam tip: For MetaMask, be ready to explain wallet address, private key, seed phrase, test network and transaction fee. For Solidity, revise contract structure, state variables, functions, structs, arrays, fallback behaviour, gas and deployment. For Hyperledger, revise the business network concept.

Scope note: The questions follow the topics and practical assignments listed in the uploaded syllabus. Some answers provide brief conceptual explanations to help with oral examination; exact implementation details depend on the code and tools used in your laboratory.